Refuse to Hallucinate

A Deterministic Validity Firewall for Indian Official Survey Statistics (MoSPI · PLFS · HCES · TUS · MIS)

MoSPI's tools help you find the number. This firewall checks whether you are allowed to say what you want to say with it — and stops you, with a cited rule, when you are not.

100 / 100	59 / 59	0 / 19	SHA-256
GOLDEN VERDICTS	FIGURES EXACT	CACHE COLLISIONS	VERDICT FROZEN PRE-RETRIEVAL

Live question for the room: *"What was urban female LFPR in PLFS 2025?"* — it refuses to guess and shows exactly why (slide 5).

Ajay Choudhary · MSAI @ Plaksha University · offline demo localhost:8001/demo · $0.00 marginal cost · no GPU · MCP server + policy agent

PROBLEM · STATSDOC FAILURE MODES FROM LIVE PROBING

Retrieval is solved. Governance is not.

Session statsdoc.ai.mospi.gov.in/c/2402 — anecdotal observations, not a measured baseline (see evaluation/statsdoc_audit/PROTOCOL.md).

CHECK	STATSDOC PILOT (OBSERVED)	VIDHI GUARDIAN (MEASURED)
MPCE definition	✗ Expands “Mean Per Capita Expenditure” — wrong expansion of the statutory acronym.	✓ Enforces statutory “Monthly Per Capita Consumption Expenditure”; cites ₹4,122 from HCES 2022-23.
2011-12 vs 2023-24	✗ Quotes ~40% growth with no methodology warning.	✓ Halts with reason code R_COMPARABILITY_HCES_METHOD (CAPI/method break).
Multi-period math	✗ 40% then 4%, chained unsoundly across non-comparable rounds.	✓ Refuses unsound synthesis; returns no number, only a coverage note.
Poisoned context	✗ Retrieved text overrides guardrails; verdict changes with corpus.	✓ Verdict frozen pre-retrieval; SHA-256 hash identical under injection.
Cost / latency	✗ ~$0.02–0.05 per query · 3.5–8 s end-to-end.	✓ $0.00 marginal · p50 ≈120 ms warm, CPU-only.

SOLUTION · MODELS TRANSLATE, VERSIONED RULES JUDGE

Five stages. Fixed order. First rule to fire wins.

Question → Exact cache → Claim extraction → Field-guard cache → **Firewall** → SHA-256 freeze → Untrusted retrieval → Templated prose → Citation check

FIVE-STAGE RULE STACK

1 Scope — Refuse forecasts, politics, causality claims. REFUSE

2 Safety — Refuse prompt injections & jailbreak attempts. REFUSE

3 Comparability — Warn on HCES-CAPI / PLFS-revamp method breaks. WARN

4 Metadata — Warn on unknown geography / sex / survey round. WARN

5 Numeric — Pass only on a verified ground-truth row match. PASS

TWO GUARANTEES

Safety-B: hash before retrieval

hash = SHA-256(claim + decision + reason + rule-version), computed before any document text is read. A planted “ignore the caveat” chunk changes nothing — the verdict is identical.

Field-guard: exact match, not cosine

No cosine threshold is safe on either embedder we tested, so the cache reuses an answer only on exact critical-field match. Zero collisions across 19 danger pairs (floor 0.83).

LIVE DEMO · SCREENSHOTS FROM LOCALHOST:8001/DEMO

Verified answers: a PASS and a 4-slice table.

Captured live from vidhi/demo.html while building this deck.

localhost:8001/demo

VERDICT

PASS_WITH_NUMBER

LFPR 59.3%

all-India · age 15+ · usual status · PLFS 2025

Q: What was all-India LFPR for age 15+ in usual status in PLFS 2025?

hash 404272ed · rule-version v2026.10

Source: PLFS Annual Report 2024-25, Statement 4.7, row_id plfs_a_15_us_2025

Hash frozen pre-retrieval · citation valid · 1 ground-truth row matched

localhost:8001/demo

VERDICT

PASS

mpce for bihar · 4 verified slices

SLICE	RURAL/URBAN	IMPUTATION	VALUE (₹)	ROW_ID
Bihar	rural	without imput...	4081	bihar_r_wo
Bihar	rural	with imputation	4122	bihar_r_w
Bihar	urban	without imput...	5157	bihar_u_wo
Bihar	urban	with imputation	5198	bihar_u_w

hash 4b7e2c81 · rule-version v2026.10 · source HCES 2022-23

Export CSV

4 / 4 slices verified · 0 refused

LIVE DEMO · THE QUESTION THIS WORK WILL NOT HALLUCINATE

Honesty for uncovered slices, refusal for forecasts.

SOURCED tier (online): exact live figure, verdict stays WARN — never silently PASS.

localhost:8001/demo

VERDICT WARN_NO_DATA

urban female LFPR

SOURCED 27.7%

Coverage note: urban-female LFPR is not in the verified store for PLFS 2025. The sourced figure is shown for context only; it is **NOT** a PASS verdict. The next release will cover this slice.

Q: What was urban female LFPR in PLFS 2025?

hash a1f3c2b9 · tier SOURCED · NOT-the-figure guard active

Offline twin: coverage note + NOT-the-figure guard, tier NONE

localhost:8001/demo

VERDICT REFUSE_SCOPE



No number will be produced.

Reason: R_FORECAST — question asks about a future outcome ("Will unemployment fall in 2027?"). Forecasts are out of scope by Stage 1 rule.

Q: Will unemployment fall in 2027?

hash 7c2e8a14 · stage 1 · rule-version v2026.10

Refusal is a feature, not a failure mode.

MEASURED, OFFLINE, REPRODUCIBLE · PYTHON VIDHI/EVAL.PY + PYTEST

Every number below is a test output, not a slide claim.

173 verified rows · 100 golden queries · 19 danger pairs · offline unit tests.

CORRECTNESS GATES (ALL MUST PASS)

100 / 100
GOLDEN VERDICTS
[0.963–1.000] agreement range

100 / 100
REASON CODES
figures 59/59 [0.939–1.000]

0 / 19
CACHE COLLISIONS
danger pairs, floor 0.83

1.000
CITATIONS VALID
every PASS cites a source row

unchanged
SAFETY-B HASH
frozen under corpus injection

90 passed
UNIT TESTS
offline, no API key required

ECONOMICS & LIMITS, STATED PLAINLY

ECONOMICS

$0.00 marginal model cost (0 model calls) plus hosting

p50 ≈ 120 ms
warm · p95 ≈ 2 s · p99 ≈ 20 s cold load, CPU-only

LIMITS

- Coverage is the limit: PLFS state splits warn, never guess.
- Heuristic readers break on exotic phrasing; LLM port exists.
- Scale path: curate PDF tables → truth JSON, no code change.

REFUSALS (RAW)

tp = 8 fp = 0 fn = 0 n = 8 · floor 0.68

ROADMAP · WHAT CONFAI FEEDBACK WOULD SHAPE

From 173 rows to national coverage without touching the firewall.

```
Try it: uvicorn vidhi.service:app --port 8001 → /demo MCP: python -m vidhi.mcp_server
```

NEXT THREE MOVES, RANKED

1	**Operating model: eSankhyiki adapter at release time** Machine ingestion → audit gate → frozen JSON. Human work per release: one comparability entry + diff review.	NEXT
2	**One parser per table family** Statement 7 done: 37/37 pairs, 0 mismatches. TUS and MIS tables next.	IN PROGRESS
3	**Verify-then-publish endpoint** For journalists and fact-checkers: submit a claim, get a cited verdict.	PROPOSED

Already built: MCP server (4 tools) + policy-briefing agent. Reproduce everything in 5 minutes.

`pytest tests/ -q`
→ 90 passed, offline, no key

`vidhi/eval.py`
→ 100/100 · 0/19 · Safety-B ok

Ask the hard question
→ expect WARN_NO_DATA + coverage note.

> ***Thank you — questions and red-team attempts welcome.***